

ENCN423 / 627 Formula Sheet

$$P = P_{max} \times (1 + (T_{cell} - T_{STC}) \cdot k_T)$$

$$E_{nominal} = \frac{E_{usable}}{DoD}$$

$$Q_{nominal} = \frac{E_{nominal}}{V_{nominal}}$$

where Q is charge

$$I_{max} = C_{rate} \times Q_{nominal}$$

$$P_{max} = I_{max} \times V_{nominal}$$

$$LCOF = \frac{CAPEX \cdot crf + OPEX + \sum (LCOI_n \cdot Input_n)}{Output_{fuel}}$$

$LCOF$ – Levelized Cost of Fuel.

$LCOI_n$ – Levelized Cost of Input n (e.g., $LCOE$, $LCOH_2$, cost of CO_2 , heat, etc.).

$$crf = \frac{r \cdot (1 + r)^{Lifetime}}{(1 + r)^{Lifetime} - 1}$$